CbcSolver: Class Architecture Overview

coin-or/Cbc — next branch — src/CbcSolver.hpp / src/CbcSolver.cpp

What is CbcSolver?

CbcSolver is the top-level driver class for the Cbc MIP solver executable and library entry point, encapsulating the entire solve pipeline — parameter initialization, command-queue parsing, LP relaxation solves, preprocessing, cut/heuristic configuration, branch-and-bound search, post-processing, and result reporting — that used to live in the free functions CbcMain0/CbcMain1, now thin backward-compatible wrappers delegating to initialize() and run().

Historically this logic was not even a class method: it lived in the free function CbcMain1 (paired with the smaller CbcMain0), a single ∼10,000-line function built around one giant switch — still exactly this shape today on stable Cbc branches. A first refactor revived the class and moved CbcMain1's body into it verbatim as CbcSolver::run() (CbcMain0/CbcMain1 remain as thin wrappers). Further refactors (removing dead #ifndef JJF_ONE / #if 0 blocks, splitting the branch-and-bound case into six cooperating private methods, dropping the legacy -miplib/-unitTest harness) then extracted coherent phases into the named methods below, reducing run() itself to a slim dispatcher. Local variables that needed to survive across phases were promoted to private data members (see §5). Most recently, three of the finer-grained **actions** used inside that pipeline — solving an LP relaxation, bound propagation, and clique strengthening — were promoted from private/inline/free-function code into public, individually-callable, solver-parameterized methods (see §2), so they can be driven directly on an arbitrary OsiSolverInterface/OsiClpSolverInterface instead of only on this CbcSolver's own model_.solver() — e.g. from the C interface.

1 Public API surface

- **Construction & lifecycle:** default / from-solver / from-model / copy constructors, operator=, destructor (lines 1997–2647); initialize() sets up signal handlers and default parameters (replaces CbcMain0).
- **High-level solve:** solve(file), importModel(file), solveLp(method) — convenience one-call wrappers that chain initialize() + run() or solve the LP relaxation directly.
- **Result accessors:** status(), objectiveValue(), bestBound(), bestSolution(), numCols(), nodeCount(), iterationCount(), hasSolution() — valid after solve()/run() returns.
- **Run (replaces CbcMain1):** run(argc,argv,...) delegates to run(deque,...), the primary command-queue–driven entry point and the root of the pipeline in §3.
- **Individual solve actions (§2):** applyLpMethod(solver), strengthenBounds(solver), strengthenCliques(solver,...) — the LP-relaxation solve, bound-propagation, and clique-strengthening actions used by the pipeline below, each independently callable and each accepting an optional target solver (defaulting to model_.solver()).
- **Heuristic / cut-generator configuration:** configureHeuristics(), configureCutGenerators() — public wrappers exposing internal BAB setup so callers can invoke it directly on a model without going through run().
- **User extensions:** addUserFunction(), setUserCallBack(), addCutGenerator() register CbcUser / CbcStopNow / CglCutGenerator plug-ins.
- **Accessors:** model(), babModel(), parameters(), intValue() / setIntValue(), doubleValue() / setDoubleValue(), originalSolver(), originalCoinModel(), getStatistics(), and similar simple getters/setters over the state in §5.

2 Individual solve actions

Three actions that used to be entangled — inline, duplicated across call sites, or reachable only as a private/free-function helper on model_.solver() — are now public CbcSolver methods that each take an optional target solver parameter (nullptr defaults to model_.solver()), so any of them can be triggered directly on an externally-owned OsiClpSolverInterface while still honouring every LP/bound-propagation/clique option configured on this CbcSolver (CbcParameters, ClpParameters):

- applyLpMethod(OsiClpSolverInterface *targetSolver=nullptr) — the unified root-LP-relaxation solver: bound propagation (→ strengthenBounds()) → clique merging "before" (→ strengthenCliques()) → model-level LP settings (PSI positive-edge, objective scaling, vector mode) → LP racing (-racingLP) → full ClpSolve with every user-settable option (-lpMethod auto/dual/primal/barrier, presolve, crash/idiot/sprint, SLP, dualize, time limits). Used internally by solveInitialLp() (root LP of the BAB pipeline) and directly for the SOLVECONTINUOUS (-initialSolve) action; exposed publicly so the exact same, fully-configured solve can be driven on any solver instance.
- strengthenBounds(OsiSolverInterface *solver=nullptr) — runs the currently configured bound propagation (-boundPropLevel: singletons / fixpoint / MILP-based, up to -boundPropMaxRounds rounds via CbcBoundPropagation), falling back to legacy singleton-only tightening when propagation is off but -singletonBounds is on. Shared by applyLpMethod() (step 1) and the standalone BOUNDPROP action (runBoundPropagation()), which previously duplicated the same level-mapping and CbcBoundPropagation::run() call inline.
- strengthenCliques(solver, mode, strengthenMode, ...) — (re)builds the conflict graph of the target solver and strengthens set-packing/partitioning cliques against it (extend/dominate), resolving the LP afterwards only in "after" mode. Thin CbcSolver-level wrapper filling in this solver's own model/handler/mipStart_ context around the lower-level free function that does the actual conflict-graph work; used by applyLpMethod()'s "before" step and by babPostPreprocessCleanup()'s "after" step, replacing what used to be two independent call sites each re-deriving the same handler/model/mipStart arguments.

As part of exposing these, a dead-code duplicate was removed from solveInitialLp(): a second clique-strengthening pass and a second model_.initialSolve() call, both gated on lpMethodResult > 0 — a condition that can never hold, since applyLpMethod() only ever returns -1 or 0 and already performs both the clique pass and the LP solve internally.

3 The branch-and-bound pipeline

run() dispatches on the current CbcParam command. Besides simple one-shot actions (§4), the BAB command drives the solver's core pipeline, implemented as six cooperating private methods called in sequence (see the full diagram on the next page):

1. babSetupAndRootLp() — cutoff sign flip, legacy OsiSolverLink path, solves the root LP via solveInitialLp() → applyLpMethod().
2. babConfigureBabModel() — (re)builds babModel_ from model_, configures the conflict-graph branching ranker (RankerConfig), tunes factorization frequency.

3. `preprocess()` — runs `CglPreProcess` when preprocessing is enabled.
4. `babPostPreprocessCleanup()` — cgraph/clique-strengthening cleanup (via `strengthenCliques()`), post-preprocess bound tightening.
5. `babConfigureSearchModel()` — knapsack expansion, cost-based priorities, and invokes `configureHeuristics()` / `configureCutGenerators()`.
6. `babExecuteSearchAndPostprocess()` — SOS/priority setup, the actual `babModel_->branchAndBound()` call, search-statistics bookkeeping, and finally `postprocess()` to translate the solution back and report results.

Each phase returns a status code consumed by `run()` to decide whether to continue, `break` the `BAB` case, or `return` from `run()` outright — preserving the exact control flow of the original monolithic function.

4 Other `run()` actions

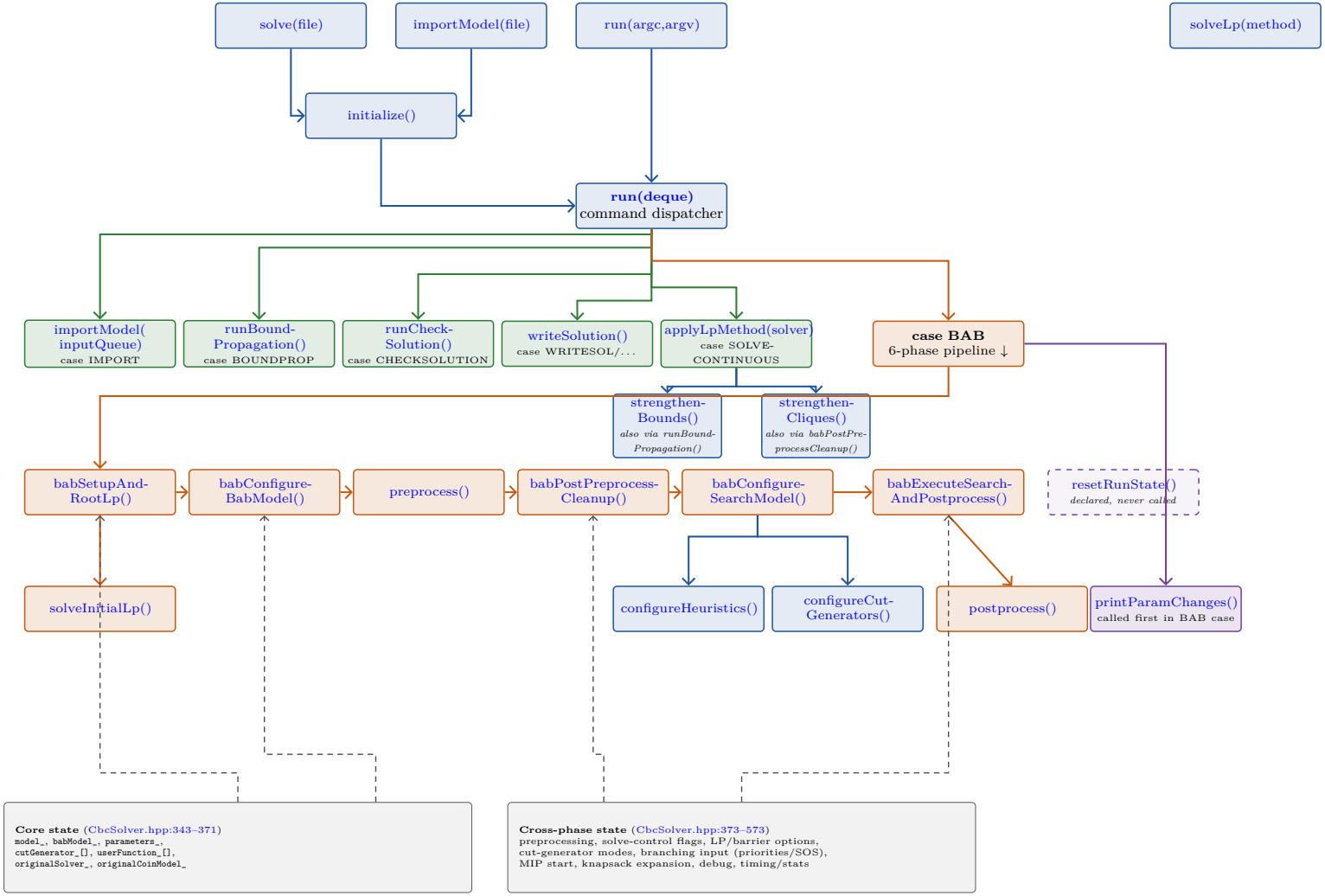
- **IMPORT:** private `importModel(inputQueue,...)` reads an MPS/LP/GMPL file.
- **SOLVECONTINUOUS:** `applyLpMethod()` solves the LP relaxation only.
- **BOUNDPROP:** `runBoundPropagation()` runs standalone bound propagation (thin CLI wrapper around `strengthenBounds()`).
- **CHECKSOLUTION:** `runCheckSolution()` writes a solution-validation report.
- **WRITESOL / PRINTSOL / ...:** `writeSolution()` writes solutions in various formats.
- **Bookkeeping:** `printParamChanges()` prints a parameter-change banner at the top of the `BAB` case; `resetRunState()` is declared/defined but currently *never called* (orphaned — dashed outline in the diagram).

5 State organization

Private data falls into two groups. **Core state** (`model_`, `babModel_`, `parameters_`, `cutGenerator_`, `userFunction_`, ...) existed in the original class. **Cross-phase state** (lines 373–573) — preprocessing (`process_`, `saveSolver_`), solve-control flags, LP/barrier options, cut-generator modes, branching input (priorities/SOS/pseudo-costs), MIP start, knapsack-expansion buffers, debug, lot-sizing, and timing/statistics — used to be *local variables* inside the monolithic `run()`. Promoting them to member fields is what let `run()` be split into the cooperating methods of §3: each extracted method reads/writes the subset of this state it needs directly, instead of receiving dozens of by-reference parameters.

Colour legend used on the diagram (next page; node fill and arrow colour share the same code, so each call arrow is coloured like the method it leaves): public API BAB pipeline (§3) other `run()` actions bookkeeping helpers state / data structures (dashed = data reads/writes)

CbcSolver — method call graph & state ownership (node labels are hyperlinks to `coin-or/Cbc@next`)



Arrows show call/return relationships within the modularized `BAB` pipeline (§3); dashed grey arrows indicate representative reads/writes of the shared state panel on the left (most pipeline methods touch several state groups — only the busiest links are drawn to keep the diagram readable). All labels link to the corresponding line in `coin-or/Cbc`, branch `next`.